Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/23/2024
PI Name:	Panontin, Tina Ph.D.		
Project Title:	Human Systems Integration Architecture Needs Analysis: Anomaly Response Analysis Mapping to Operations Beyond Low-Earth Orbit		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2022	**End Date:**	10/01/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Shu-Chieh (San Jose State University Research Foundation Inc) McTigue, Kaitlin M.S. (NASA Ames Research Center) Vera, Alonso Ph.D. (NASA Ames Research Center) Parisi, Megan (NASA Ames Research Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
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Task Description:	Distance from Earth will reduce ground support for the execution of complex nominal and off-nominal safety-critical operations during future deep space exploration missions. The likelihood of a loss-of-mission outcome due to delayed communications with Mission Control during procedure execution for Artemis lunar surface operations is estimated to be greater than 10% for a 30-day mission. For a Mars mission, the likelihood of high consequence problems of uncertain origin occurring during transit – one that requires the crew to act alone to solve — is high (exceeding 50% based on historical trends). This is characterized as a Human Systems Integration Architecture (HSIA) risk because mitigation depends on identifying onboard capability requirements and crew roles and responsibilities necessary to enable effective and efficient crew responses to the aforementioned situations. Previous needs analysis revealed the outline of typical anomaly responses from historical incidents, which are mostly driven by mission control on the ground. To be able to identify onboard capability requirements for future vehicles, higher-level resolution analysis is needed from those incidents. The purpose of the present work is to more precisely characterize the anomaly response processes used by the ground control team currently. These processes are then analyzed to determine how they might work on the lunar surface or during Mars transit, when the crew is in the lead with the ground supporting, to reveal those most negatively impacted by communication latencies. Specifically, two International Space Station (ISS) incidents are analyzed in detail, first to examine the processes as they occurred in real time during ISS operations and then as they are projected to occur during a Mars transit. Onboard capabilities needed to support the crew in performing urgent problem solving when communication with the ground is delayed are identified to inform the design of vehicle/habitats, interfaces, procedures, and training for missions beyond low Earth orbit (BLEO). This work also outlines criteria for devising problem-solving tasks for use in mission analogs to assess the utility and impacts of onboard support technologies. Preliminary results suggest that lengthy communication latencies (20 min one-way) will significantly disrupt the anomaly resolution process as it is currently performed for low Earth orbit (LEO) missions. Assignment of problem-solving roles and responsibilities will likely need to be adjusted with time-critical decision-making and vehicle management shifted from the ground team to the onboard crew with increasing distance from Earth. Crew will need onboard systems to help them understand the context for telemetry, engineering, and safety data to effectively detect, triage, and respond to malfunctions. Under certain conditions, commanding and control will need to move onboard: with reduced situational awareness and delayed feedback, the ground loses the insight needed to ensure safe execution during off-nominal operations. Hypothesis generation and testing needed for urgent troubleshooting may also need to be performed onboard as iterative processes between the ground team and crew under comm delayed conditions are inefficient in general and ineffective for time-constrained responses. Early results indicate both asynchronous collaboration with the ground and onboard human-machine teaming are essential for successful anomaly resolution in deep space. The proposed work directly addresses Gap HSIA-401, specifically what capabilities onboard intelligent systems will need to provide to support autonomous crew operations.
Rationale for HRP Directed Research:	This proposal qualifies for a directed study due to the urgent, time-sensitive need to develop HSIA risk characterization research for Artemis missions and beyond. Upcoming Artemis missions can provide important insights into the HSIA risk characterization, and the schedule for the Artemis II mission in particular results in a compressed timeline to prepare research and collect data. Timely data analysis can inform future Artemis missions as well as missions to Mars. In order to meet the HRP milestones for Lunar and Mars transit missions, and to establish a baseline human performance threshold, this work must be directed due to insufficient time for solicitation. Additionally, the directed nature of this study allows the HSIA Discipline Scientist to leverage previously collected data that may inform solutions.
Research Impact/Earth Benefits:	Our research program and associated deliverables will contribute to reducing the Earth Independent Human-Systems Operations (EIHSO) risk, formerly known as Human-Systems Integration Architecture (HSIA) risk. They will also inform gaps in knowledge relevant to maintaining crew health and performance on future long duration exploration missions beyond low-Earth orbit (LEO).
Task Progress:	Final presentation of this project, entitled “Human Systems Integration Architecture Needs Analysis: Anomaly Response Analysis Mapping to Operations Beyond Low-Earth Orbit,” was delivered to the Human Factors and Behavioral Performance Element on September 20, 2023.
Bibliography Type:	Description: (Last Updated: 01/29/2025)
Abstracts for Journals and Proceedings	McTigue K, Panontin T, Parisi M, Vera A, Wu S-C, Holden K, Robertson I. "Simulating the unexpected: Assessing human-systems resilience." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Buckland DM, Parisi M, Mctigue K, Wu S, Panontin T, Vos G, Petersen D, Vera A. "NASA’s identified risk of adverse outcomes due to inadequate human systems integration architecture." 13th International Conference on Applied Human Factors and Ergonomics (AHFE 2022), New York, New York, July 24-28, 2022. Abstracts. 13th International Conference on Applied Human Factors and Ergonomics (AHFE 2022), New York, New York, July 24-28, 2022. , Jul-2022
Articles in Peer-reviewed Journals	Valinia A, Folta D, Hughes K, Hatten N, Vera A, Stone L, Parisi M, McTigue K, Panontin T. "Risk trade-space analysis for safe human expeditions to Mars." Acta Astronaut. Oct 2023;211:192-99. https://doi.org/10.1016/j.actaastro.2023.05.039 , Oct-2023
Books/Book Chapters	Panontin T, Parisi M, Wu S-C, McTigue K, Vera A. "Human-systems integration challenges." in "In: Systems Medicine for Human Spaceflight." Ed. M. Schelhamer, E. Antonsen. Hackensack, NJ: World Scientific Publishing Co. Pte. Ltd, 2024. p. 151-81., May-2024